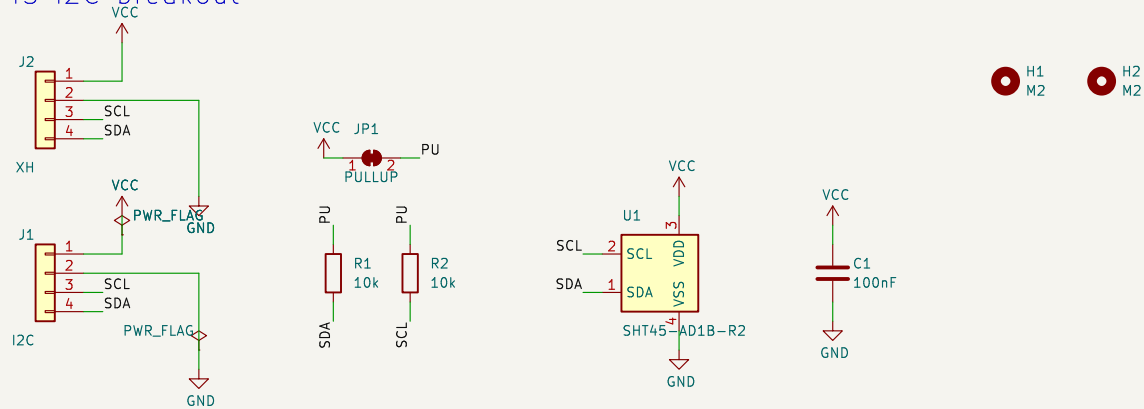


SHT45 I2C breakout



J1 (2.54 mm header) / J2 (JST XH 2.50 mm): 1=VCC 2=GND 3=SCL 4=SDA
VCC = 1.08 ... 3.6 V (NOT 5 V tolerant!)
I2C address 0x44 (SHT45-AD1B), up to 1 MHz (Fm+)

R1/R2: 10k pull-ups as in datasheet Fig.1 ($R_p \geq 390R @ VDD \geq 1.62V$).
JP1 (bottom side) is bridged by default. Cut the trace between its pads
to disconnect both pull-ups when the bus already has them.

C1: 100nF placed directly at U1 VDD/VSS (datasheet Fig.1).
U1: die pad NOT soldered and no copper under the sensor (datasheet 5.3).
U1 sits on a narrow PCB tongue for thermal decoupling (Sensirion Design Guide ch.3).

Board 16.0 x 19.8 mm, 2 layers, 2x M2 holes
Sensirion SHT45 humidity & temperature sensor, I2C

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File: sht45_breakout.kicad_sch

Title: SHT45 Breakout

Size: A4 Date: 2026-09-29

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